Sheet 15 Tue. 24 Nov. 2026 Irrational numbers: $\frac{1}{3}$ is not a decimal, $\sqrt{2}$ is irrational

1. Warm-up · Flash Write $\sqrt{12}$ and $\sqrt{50}$ in the form $a\sqrt{b}$ with b an integer as small as possible.

2. Core · Sorting Among $0.333\ldots; \sqrt{9}; \sqrt{2}; \pi; \frac{22}{7}$, which ones are rational?

3. Core + · Reasoning Suppose $\sqrt{2} = \frac{p}{q}$ with $\frac{p}{q}$ irreducible, hence $p^2 = 2q^2$. What can be said about the parity of p, then of q ? What contradiction?

4. Review · Table Values 1; 2; 3; 4 with frequencies 5; 8; 4; 3. Give the total frequency, the relative frequency of the value 2 and the mean.

Value	1	2	3	4
Count	5	8	4	3

5. Running thread · Flash Give the coordinates of the reflection of $A(3;-2)$ in the y-axis.

Sheet 16 Thu. 26 Nov. 2026 Intervals of $\mathbb{R}$: notation and representation

1. Warm-up · Flash Compare -3 and -5 then $-\frac{1}{2}$ and $-0{,}6$.

2. Core · Method Write as an interval: $-2 \leqslant x < 5\,;\, x > 3\,;\, x \leqslant 0$.

3. Core + · MCQ The intersection $]-\infty;2[\cap [-1;4]$ is:

A. $]2;4]$ **B.** $]-\infty;4]$ **C.** $[-1;2[$ **D.** $[-1;2]$

4. Review · Table Travel times (min).

Class	$[0;10[$	$[10;20[$	$[20;30[$
Count	4	9	7

What are the total frequency, the modal class and the frequency of $[10;20[$?

5. Running thread · Problem One bus comes every 15 min, another every 20 min. They meet at 8:00. At what time do they meet again?

Sheet 17 Mon. 30 Nov. 2026 Absolute value and distance between two real numbers

1. Warm-up · Flash What is the distance between -3 and 5 on the number line?

2. Core · Flash Calculate $|-7|$, $|3-8|$ and $|\pi - 4|$.

3. Core + · Method Solve $|x-3| = 2$.

4. Review · Formula $f(x) = x^2 - 3$. Give the image of -2 and the preimages of 6.

5. Running thread · Formula → table $f(x) = -x + 3$. Complete: $f(-1), f(0), f(2)$.

Sheet 18 Tue. 1st Dec. 2026 Inequality $|x-a| \leqslant$ r and associated interval

1. Warm-up · Flash Solve 2 $x - 3 \leqslant 5$.

2. Core · Method Solve $|x-2| \leqslant 3$ and give S as an interval.

3. Core + · Translate Express as an inequality with an absolute value: $x \in [-1;5]$ then $x \in]4;10[$.

4. Review · Problem Pack A: 6 yoghurts for 4.20 €. Pack B: 10 yoghurts for 6.80 €. Which is the better deal?

5. Running thread · Flash A disc has radius 5 cm. Give its area and its perimeter (exact values).

Functions: representations and reference functions

Sheet 19 Thu. 3 Dec. 2026 Real-valued function; domain

1. Warm-up · Flash Solve $x - 4 = 0$, then $2\ x + 6 = 0$.

2. Core · Method Domain of $f(x) = \dfrac{1}{x-4}$?

3. Core + · MCQ The domain of $h(x) = \dfrac{1}{x^2+1}$ is:

A. $\mathbb{R} \setminus \{1\}$ **B.** $\mathbb{R} \setminus \{-1\,;\,1\}$ **C.** $\mathbb{R}$ **D.** $\mathbb{R}^*$

4. Review · Table Is this a table of proportional values?

x	2	5	8
y	6	15	24

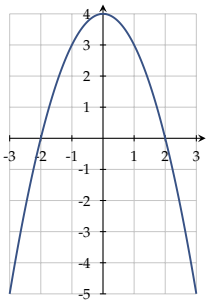
5. Running thread · Reading code What is returned by $f(4)$?

```
def f(x):
    return 3*x - 2
```

Sheet 20 Mon. 7 Dec. 2026 Graph with equation y $= f(x)$; images and preimages

1. Warm-up · Flash $f(x) = -x^2 + 4$. Calculate $f(0)$, $f(-2)$ and $f(3)$.

2. Core · Graph Graph of f :



Read $f(1)$ and the preimages of 0.

3. Core + · Reflection For this function, does the equation $f(x) = 5$ have solutions?

4. Review · Flash Urn: 5 red balls, 3 blue, 2 green. Probability of drawing a blue one? A non-red?

5. Running thread · Method Solve in $\mathbb{R}$: $4 - 3x = 10$.

Sheet 21 Tue. 8 Dec. 2026 Reference functions: absolute value, square, reciprocal

1. Warm-up · Flash Calculate $|-2{,}5|$, $(-3)^2$ and $\dfrac{1}{4}$.

2. Core · Table Complete for $x = -2\,;\,-1;\,0;\,1;\,2$ the values of $x \mapsto |x|$, $x \mapsto x^2$ and $x \mapsto \dfrac{1}{x}$.

3. Core + · MCQ Which reference curve passes through $(-1;1)$, $(0;0)$, $(1;1)$ and $(2;4)$?

A. the line $y = x$ **B.** that of the reciprocal **C.** that of the absolute value **D.** that of the square

4. Review · Flash An object is enlarged by scale factor 1.5. What does a length of 8 cm become? By what number is the area multiplied?

5. Running thread · Prove Conjecture: "the sum of two consecutive integers is odd". Prove it.